

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 4.3 - Passwords, authentication, and the cloud

A real classroom situation

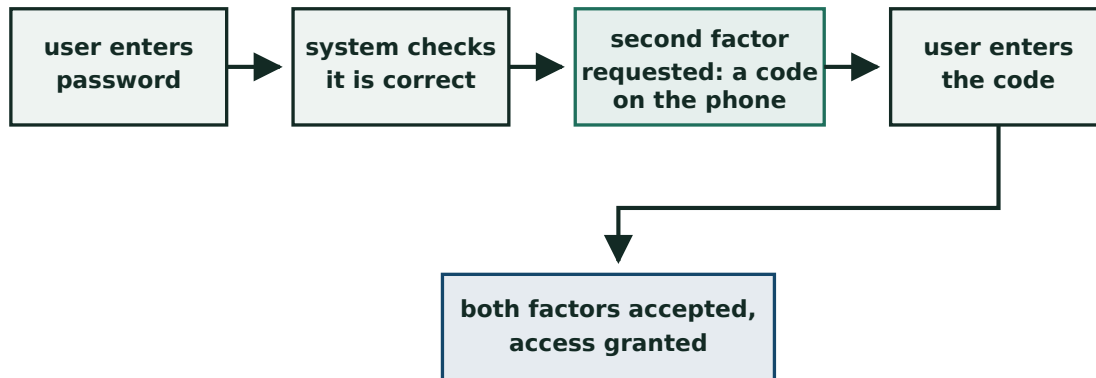
The school office computer's marks portal is protected by the password "school123". It meets the portal's rule - eight characters, one word, easy for everyone on the staff to remember and share. That is exactly the problem: a password everyone can share is a password everyone can use, including someone who should not. When the portal is compromised, the technician's first instruction is not a new firewall; it is a new way of thinking about secrets - and a second lock on the door that no shared word can be.

How multi-factor authentication works

Something you know, plus something you have

factor one: something you know

factor two: something you have (the phone)



A phished password alone opens nothing: the attacker still needs your phone.

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Explanation

Password management and authentication

A password works by being a secret between you and a system. Three habits keep it one. **Length and variety beat cleverness:** Bourgeois and colleagues (2019) report that the most common real-world passwords were literally "password", "123456" and "12345678" - the first things an attacker tries - and recommend at least twelve characters for a personal password, mixing letters of both cases, digits and symbols, with no dictionary words. **Never reuse a password across accounts:** one phished site should not unlock your email, your bank and the school portal. **Never share it:** an attacker who cannot guess a password will often simply ask for it (that is pretexting from Topic 4.1).

Because human memory is the weakest link, the modern answer is a password manager: an application that generates and stores a different strong password for every account, locked behind one master password you do remember. One good secret replaces forty weak ones.

Authentication is the system's act of checking that you are who you claim to be, and it can rest on three kinds of evidence: something you know (a password), something you have (a phone or a security key), and something you are (a fingerprint or face). Multi-factor authentication (MFA) combines two different kinds - a password plus a code sent to your phone, as the figure shows. Its power is simple: an attacker who phishes your password still cannot log in, because the second factor is in your pocket. Every account that matters - email first of all, because email resets everything else - should have MFA turned on.

Judging passwords and habits

Password or habit	Verdict	Why
school123	weak	dictionary word plus digits, tried first by attackers
Shakeel2010	weak	a name plus a birth year, guessable from your footprint
Pk!9#mQ2vLx	strong	twelve characters, mixed types, no words
a unique password per account	strong habit	one phished site unlocks nothing else
the same password everywhere	dangerous habit	one leak opens every door

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Cloud computing basics

"The cloud" is not a place in the sky; it is other people's computers, rented by the minute. The standard definition, from NIST, runs: cloud computing is "a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources... that can be rapidly provisioned and released with minimal management effort or service provider interaction" (Mell and Grance, 2011). The definition names five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service - you get what you need when you need

it, over the network, from a shared pool, scaling up and down, paying for what you use.

For a school, the practical translation is threefold. **Storage:** files kept in the cloud (a drive, a backup service) live on the provider's machines, copied across data centres - which is why a stolen laptop no longer means lost records.

Services: email, documents, and learning platforms that run in the browser are cloud services, nothing installed on the lab machines. **Someone else's responsibility, someone else's rules:** the provider runs the servers, but you have handed them your data, so the provider's reputation, terms and location matter. The cloud is a trade: convenience and durability in exchange for trust. Topic 4.4 takes up what that trust should require.

Activity: Password workshop

Work in pairs. About 20 minutes.

1. Your teacher writes five passwords on the board (from the second figure's list, plus two new ones). For each, agree a verdict - weak, strong, or dangerous habit - and one clause of reasoning.
2. Together, invent a strong password you could actually remember, using a method (not a word): for example, the first letters of a sentence only you know, with a number and symbol added. Write the method, not the password itself.
3. List which of your own accounts currently has MFA turned on, and mark the two that most need it.
4. Discuss: what would you tell the school office that currently shares one password among five people? Propose a workable alternative in two sentences.

Check your understanding

1. Name the three kinds of authentication evidence, with one example of each. (*Remember*)
2. Explain why "school123" is a weak password even though it has eight characters. (*Understand*)

3. Your email account is phished but has MFA enabled. Explain what happens when the attacker tries to log in. (*Apply*)
4. State the NIST definition of cloud computing in your own words, naming two of its five characteristics. (*Understand*)

Summary

A password is a secret, and the habits that keep it one are length and variety, no reuse, and no sharing - with a password manager doing the remembering. Authentication can rest on something you know, have, or are, and multi-factor authentication combines two kinds, so a phished password alone opens nothing. The cloud is other people's computers: on-demand, networked, pooled, elastic, metered - convenient and durable, at the price of trusting the provider with your data.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can judge a password's strength and explain my verdict.
- ☐ I can explain how MFA works and why it defeats a phished password.
- ☐ I can explain what the cloud is and name its five characteristics.

Try this at your practicum school

Ask the office (politely) whether the school's portal or email uses any second factor today, and how many people share the main password. Write two sentences: the current practice, and the single change you would recommend first.

Summative task

The school office shares one password for the marks portal among five staff, and the head teacher keeps the only backup drive in the same cupboard as the

computer. In one page, recommend a security upgrade: what to change about the passwords and why, whether and how to add a second factor, and what to do about the backup - connecting each recommendation to a threat from Topic 4.1.

Marking guide (10 marks):

- The password problem diagnosed with its real risks (sharing, weakness, reuse): 3 marks.
- The MFA recommendation explains the mechanism, not just the name (Analyse): 3 marks.
- The backup recommendation addresses the same-location failure: 2 marks.
- Each recommendation is tied to a named threat: 1 mark.
- Clear, one page: 1 mark.

Further reading

- Mell, P., & Grance, T. (2011). *The NIST Definition of Cloud Computing* (Special Publication 800-145). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-145>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 6. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- National Cybersecurity Alliance. *Stay Safe Online*. <https://staysafeonline.org/>

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